

A longitudinal data analysis interpretation of credibility models

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Abstract

In this paper, we develop links between credibility theory in actuarial science and longitudinal data models in statistics. Our primary contribution to actuarial science is to demonstrate that many additive credibility models can be expressed as special cases of the longitudinal data model. We, thereby, unify the many existing credibility models with this framework. In addition, a longitudinal data interpretation suggests additional models and techniques that actuaries can use in credibility ratemaking. We also apply standard statistical software, which has been developed to analyze longitudinal data models, to the private passenger automobile data of Hachemeister [Hachemeister, C.A., 1975. Credibility for regression models with applications to trend. In: Kahn, P.M. (Ed.), *Credibility: Theory and Applications*. Academic Press, New York, pp. 129–163]. ©1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

In actuarial science, a fundamental problem is to predict future claims of a risk class, given past claims of that and related risk classes. This is the credibility ratemaking problem. In statistics and econometrics, *longitudinal data models* have been developed to describe cross-sectional data that are tracked over time and to predict future observations. In this paper, we develop links between credibility theory and longitudinal data models.

Longitudinal data models have been developed in statistics and econometrics. In statistics and the life sciences, early researchers such as Potthoff and Roy (1964) used versions of these models to study the growth of a group of children over time. Recent monographs such as Diggle et al. (1994) and Hand and Crowder (1996) survey a broad range of applications from the agricultural and biomedical sciences.

In econometrics and the social sciences, longitudinal data models are commonly referred to as *panel data models*. This nomenclature refers to applications that study collections of people, called panels, in large surveys conducted on populations of interest. Panel surveys have the same structure as longitudinal data in that individuals in a panel are surveyed on repeated occasions. The econometric literature, as in Hsiao (1986) and Baltagi (1995), describes

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many applications wherein cross-sections of people, firms, governmental units and other entities of interest are followed over time.

In this paper, we show how longitudinal data models can be applied to the credibility ratemaking problem. To illustrate, one can view an insurer's collection of risk classes as generating cross-sectional data in any one policy period. Then, an insurer observes the claims for those risk classes over several periods. Further, it may be that there are explanatory characteristics of the risk class that can be used to predict claims. Thus, one can model claims across risk classes and across time by longitudinal data models. Longitudinal data models allow us to handle categorical variables needed for classification models as well as continuous variables, such as trend variables in Hachemeister's (1975) model. Further, within the longitudinal data model framework we can easily handle serial correlation as another method for modeling time trends.

This paper is similar in flavor to Norberg (1986) who used special cases of the statistical mixed linear model to investigate multi-stage, nested classification, credibility models. We use another version of the mixed linear model, the longitudinal data model, to investigate other linear credibility models. As with Norberg's work, we find that the connections that we develop are particularly useful for estimating components of the variance, a subject well developed in the statistical literature.

By demonstrating that many important credibility models can be viewed in a longitudinal data framework, we restrict our consideration to certain types of credibility models. Specifically, the longitudinal data models accommodate only unobserved risks that are additive. Thus, we do not address models of nonlinear random effects that have been investigated in the actuarial literature; see, for example, Taylor (1977) and Norberg (1980). Section 6 describes additional types of credibility models that cannot be expressed directly within the longitudinal data framework. We assume that the readers are familiar with credibility theory. For readers who would like more background in credibility theory, please refer to Goovaerts and Hoogstad (1987), Dannenburg et al. (1996) and Klugman et al. (1998).

By expressing credibility ratemaking applications in the framework of longitudinal data models, actuaries can realize several benefits: (1) Longitudinal data models provide a wide variety of models from which to choose. (2) Standard statistical software makes analyzing data relatively easy. (3) Actuaries have another method for explaining the ratemaking process. (4) Actuaries can use graphical and diagnostic tools to select a model and assess its usefulness.

In Section 2, we describe longitudinal data models and show how many well-known credibility models fit into the longitudinal framework. In Section 3, we review the general prediction problem and its application to longitudinal data models. Then, we apply the results to the credibility models considered in Section 2. Prediction in a general regression setting is a concept first discussed by Goldberger (1962), an econometrician. Goldberger's predictors, now termed best linear unbiased predictors, or BLUPs, are widely used in animal breeding and other agricultural applications. Because prediction of future claims is a key element of credibility theory, we review the details from a scattered literature so that this article is self-contained.

In Section 4, we discuss parameter estimation. Because credibility estimators are often based on estimators of variability, we focus on estimation of variance components. In addition to maximum likelihood, we describe a restricted version, called REML, that was developed in the 1970s as well as Rao's (1970) minimum variance unbiased quadratic norm estimators. Both are standard features in statistical software packages. These estimation routines supplement the parameter estimation procedures in the actuarial literature described by Dannenburg (1994) and DeVyllder (1981).

In Section 5, we demonstrate how to apply longitudinal data methods to the private passenger automobile data from Hachemeister (1975). In Section 6, we provide a summary and some concluding remarks.

2. Longitudinal data models

In this section, we first describe the longitudinal data model and provide an estimator of one of the model's (vector) parameters via generalized least squares estimation. Then, we show that several well-known credibility models can be expressed as longitudinal data models.

2.1. Model specification and generalized least squares estimation

The longitudinal data model is

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + \mathbf{z}_{it}\boldsymbol{\alpha}_i + \varepsilon_{it}. \quad (1)$$

Subject i is observed on T_i occasions, where $i = 1, \dots, n$ and $T_i \leq T$, the maximal number of time periods. The K explanatory variables are $\mathbf{x}_{it} = (x_{it1}, x_{it2}, \dots, x_{itK})$, a known row vector of dimension $1 \times K$, and the corresponding unknown non-subject-specific parameters are $\boldsymbol{\beta} = (\beta_1, \dots, \beta_K)^t$. An additional q known explanatory variables are $\mathbf{z}_{it} = (z_{it1}, z_{it2}, \dots, z_{itq})$ a row vector of dimension $1 \times q$. The subject-specific random parameters are $\boldsymbol{\alpha}_i = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iq})^t$. The error, or noise, term is ε_{it} .

A matrix form of this model is

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\boldsymbol{\alpha}_i + \boldsymbol{\varepsilon}_i. \quad (2)$$

Here, $\mathbf{y}_i$ is the $T_i \times 1$ vector of observations for the i th subject, $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{iT_i})^t$. Similarly, $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \dots, \varepsilon_{iT_i})^t$. Further, $\mathbf{X}_i$ is a $T_i \times K$ matrix of explanatory variables, $\mathbf{X}_i = (\mathbf{x}_{i1}^t, \mathbf{x}_{i2}^t, \dots, \mathbf{x}_{iT_i}^t)^t$ and $\mathbf{Z}_i$ is a $T_i \times q$ matrix of explanatory variables, $\mathbf{Z}_i = (\mathbf{z}_{i1}^t, \mathbf{z}_{i2}^t, \dots, \mathbf{z}_{iT_i}^t)^t$.

The responses between subjects are independent, yet we allow for serial correlation and weighting by assuming that $\text{Var}(\boldsymbol{\varepsilon}_i) = \mathbf{R}_i$. In the multi-dimensional case, we write Var to denote the variance–covariance matrix that contains variances on the diagonal and covariances off the diagonal. Also, $E(\boldsymbol{\varepsilon}_i) = \mathbf{0}_{1 \times T_i}$, in which $\mathbf{0}_{r \times s}$ is an $r \times s$ matrix with all 0's. When it is clear from the context, we write $\mathbf{0}$ for $\mathbf{0}_{r \times s}$. Further, we assume that the subject-specific effects $\{\boldsymbol{\alpha}_i\}$ are independent and identically distributed with mean zero and variance–covariance matrix $\mathbf{D}$, a $q \times q$ positive definite matrix. Subject-specific effects and the noise term are assumed to be uncorrelated; that is, $\text{Cov}(\alpha_{jr}, \varepsilon_{it}) = 0$, for all i, j, r , and t . Thus, the variance–covariance matrix of each subject can be expressed as

$$\text{Var}(\mathbf{y}_i) = \mathbf{Z}_i\mathbf{D}\mathbf{Z}_i^t + \mathbf{R}_i = \mathbf{V}_i.$$

We assume that $\mathbf{V}_i$ is invertible, for each i . Due to symmetry and the relation $T_i \leq T$, there are at most $T(T+1)/2$ unknown elements of $\mathbf{V}_i$. Denote this dependence of $\mathbf{V}_i$ on parameters using $\mathbf{V}_i(\boldsymbol{\tau})$. Thus, $\boldsymbol{\tau}$ is the vector of unknown parameters that determine $\mathbf{V}_i$ and are called the variance components.

Example 1 (Balanced Bühlmann). Let $K = q = 1$, $x_{it} = z_{it} = 1$, and $T_i = T$, $i = 1, 2, \dots, n$. Then, expression (1) becomes

$$y_{it} = \beta + \alpha_i + \varepsilon_{it},$$

where $i = 1, 2, \dots, n$, and $t = 1, 2, \dots, T$. If y_{it} represents the claims of the i th subject in period t , then β is the grand mean of the claims over the collection of subjects (policyholders, geographical regions, occupational classes, etc.), and α_i is the deviation of the i th subject's hypothetical mean from the overall mean β . Here, the hypothetical mean is the conditional expected value $E(y_{it}|\alpha_i) = \beta + \alpha_i$. The noise term ε_{it} is the deviation of y_{it} from its hypothetical mean. In this case, $\mathbf{D} = \text{Var}(\alpha_i) = \sigma_\alpha^2$ and $\mathbf{R}_i = \text{Var}(\boldsymbol{\varepsilon}_i) = \sigma_\varepsilon^2 \mathbf{I}_T$, in which $\mathbf{I}_T$ is the $T \times T$ (multiplicative) identity matrix. See Bühlmann (1967) for an early description of this model.

The generalized least squares (GLS) estimator of $\boldsymbol{\beta}$ assumes that the variance parameters, $\boldsymbol{\tau}$, are known. Section 4 addresses the problem of estimating $\boldsymbol{\tau}$. The GLS estimator of $\boldsymbol{\beta}$ is

$$\mathbf{b}_{\text{GLS}} = \left(\sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i \right)^{-1} \left(\sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{y}_i \right).$$

Example 2 (Balanced Bühlmann, continued). An easy calculation shows that the GLS estimator of β for the balanced Bühlmann model is $b_{\text{GLS}} = \bar{y}$, the sample mean.

Table 1
Credibility and generalized least squares (GLS) estimators

Model	Notation	Credibility factors	GLS estimators($\mathbf{b}_{\text{GLS}}$)
Bühlmann	$\bar{y}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} y_{it}$	$\zeta_i = \frac{T_i}{T_i + \sigma_\epsilon^2 / \sigma_\alpha^2}$	$\bar{y}_w = \frac{\sum_{i=1}^n \zeta_i \bar{y}_i}{\sum_{i=1}^n \zeta_i}$
Bühlmann–Straub	$\bar{y}_{i,w} = \frac{\sum_{t=1}^{T_i} w_{it} y_{it}}{\sum_{t=1}^{T_i} w_{it}}$	$\zeta_i = \frac{\sum_{t=1}^{T_i} w_{it}}{\sum_{t=1}^{T_i} w_{it} + \sigma_\epsilon^2 / \sigma_\alpha^2}$	$\bar{y}_w = \frac{\sum_{i=1}^n \zeta_i \bar{y}_{i,w}}{\sum_{i=1}^n \zeta_i}$
Hachemeister (linear trend)	$\mathbf{D} = \begin{bmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{bmatrix}$ $\mathbf{W}_i = \begin{bmatrix} \sum_{t=1}^{T_i} w_{it} & \sum_{t=1}^{T_i} t w_{it} \\ \sum_{t=1}^{T_i} t w_{it} & \sum_{t=1}^{T_i} t^2 w_{it} \end{bmatrix}$	$\zeta_i = \frac{\det(\mathbf{D}\mathbf{W}_i)\mathbf{I}_2 + \sigma_\epsilon^2 \mathbf{D}\mathbf{W}_i}{\det(\mathbf{D}\mathbf{W}_i) + \sigma_\epsilon^2 \text{trace}(\mathbf{D}\mathbf{W}_i) + \sigma_\epsilon^4}$	$(\sum_{i=1}^n \zeta_i)^{-1} \left(\sum_{i=1}^n \zeta_i \mathbf{W}_i^{-1} \left[\sum_t w_{it} y_{it} \right] \right)$
Jewell	$A_i = \sum_{j=1}^{r_i} \frac{\sigma_\gamma^2 \sum_{t=1}^{T_{ij}} w_{ijt}}{\sigma_\gamma^2 \sum_{t=1}^{T_{ij}} w_{ijt} + \sigma_\epsilon^2}$ $\bar{y}_{ij,w} = \frac{\sum_{t=1}^{T_{ij}} w_{ijt} y_{ijt}}{\sum_{t=1}^{T_{ij}} w_{ijt}}$	$\zeta_i = \frac{\sigma_\mu^2 A_i}{\sigma_\mu^2 A_i + \sigma_\gamma^2}$ and $\zeta_{ij} = \frac{\sigma_\gamma^2 \sum_{t=1}^{T_{ij}} w_{ijt}}{\sigma_\gamma^2 \sum_{t=1}^{T_{ij}} w_{ijt} + \sigma_\epsilon^2}$	$\bar{y}_w = \frac{\sum_{i=1}^n \zeta_i \frac{\sum_{j=1}^{r_i} \zeta_{ij} \bar{y}_{ij,w}}{\sum_{j=1}^{r_i} \zeta_{ij}}}{\sum_{i=1}^n \zeta_i}$

2.2. Credibility theory models

In this subsection, we demonstrate that commonly used credibility models are special cases of the longitudinal data model and calculate $\mathbf{b}_{\text{GLS}}$ for each case. See Table 1 for a summary of these calculations; note that in each case, $\mathbf{b}_{\text{GLS}}$ is a weighted average of the observations with weights involving the credibility factors. In Section 3, we consider the problem of prediction in each of these models.

2.2.1. Basic credibility model of Bühlmann

As in the balanced Bühlmann example in Section 2.1, let $K = q = 1$ and $x_{it} = z_{it} = 1$, for all $i = 1, 2, \dots, n$ and $t = 1, 2, \dots, T_i$. Note that here we allow unequal numbers of observations T_i per subject. In this case, $\mathbf{D} = \text{Var}(\alpha_i) = \sigma_\alpha^2$ and $\mathbf{R}_i = \text{Var}(\epsilon_i) = \sigma_\epsilon^2 \mathbf{I}_{T_i}$. One calls σ_α^2 the variance of the hypothetical means and σ_ϵ^2 the process variance (Bühlmann, 1967). See Table 1 for the GLS estimator of the grand mean β . Note that $\mathbf{b}_{\text{GLS}}$ is a weighted average of $\{\bar{y}_i\}$ with weight $T_i / (T_i + \sigma_\epsilon^2 / \sigma_\alpha^2)$ which is the familiar form of the Bühlmann credibility factor.

2.2.2. Heteroscedastic model of Bühlmann–Straub

Continue with the example in Section 2.2.1, and change only the variance–covariance matrix of the errors $\mathbf{R}_i = \text{Var}(\epsilon_i) = \sigma_\epsilon^2 \text{diagonal}(1/w_{i1}, \dots, 1/w_{iT_i})$. By this change, we allow each observation to have a different ‘exposure’ weight (Bühlmann and Straub, 1970). For example, if a subject is a policyholder, then w_{it} measures the size of the i th policyholder’s exposure during the t th period, possibly via payroll as for workers compensation insurance. Note that $\mathbf{b}_{\text{GLS}}$, in Table 1, reduces to the expression we found in Section 2.2.1 when $w_{it} = 1$, for all i and t .

2.2.3. Regression model of Hachemeister

Using Eq. (1), assume $\mathbf{x}_{it} = \mathbf{z}_{it}$ to get a random coefficients model. Then, with $\mathbf{R}$ as in Section 2.2.2, we have the regression model of Hachemeister (1975). In Section 5, we consider his linear trend model for which $K = q = 2$, $\mathbf{x}_{it} = \mathbf{z}_{it} = (1 \ t)$. See Table 1 for the GLS estimator of β in this case.

2.2.4. Hierarchical (nested) classification model of Jewell

Suppose $y_{ijt} = \beta + \mu_i + \gamma_{ij} + \epsilon_{ijt}$, a sum of uncorrelated components, in which β is the overall expected claims, μ_i is the deviation of the conditional expected claims of the i th sector from β , $i = 1, 2, \dots, n$, γ_{ij} is the

deviation of the conditional expected claims of the j th subject in the i th sector from the sector expectation of $\beta + \mu_i$, $j = 1, 2, \dots, r_i$, and ε_{ijt} is the deviation of the observation y_{ijt} from $\beta + \mu_i + \gamma_{ij}$, $t = 1, 2, \dots, T_{ij}$, (Jewell, 1975). The conditional expected claims from the i th sector is $E(y_{ijt}|\mu_i) = \beta + \mu_i$ and the conditional expected claims of subject j within the i th sector is $E(y_{ijt}|\mu_i, \gamma_{ij}) = \beta + \mu_i + \gamma_{ij}$. If one were to apply this model to private passenger automobile insurance, for example, then the sector might be age of the insured(s) while the subject is geographical region of the insured(s). Note that one assumes with this model that the claims in region j for different ages are uncorrelated. If one believes this to be an unreasonable assumption, then a cross classification model might be appropriate; see Dannenburg et al. (1996). As an example for which this hierarchical model might be more appropriate, one could let the sector be geographical region, while the subject is the policyholder.

To write the model in terms of Eq. (2), first define $q = \max(r_1, \dots, r_n)$, and $T_i = \sum_{j=1}^{r_i} T_{ij}$. Let $\alpha_{ij} = \mu_i + \gamma_{ij}$ and $\alpha_i = (\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iq})^t$. Define $\mathbf{X}_i = \mathbf{1}$, a $T_i \times 1$ column vector of 1's. Further, let $\mathbf{Z}_{i1} = (\mathbf{e}_1 \mathbf{1}_{1 \times T_{i1}}, \dots, \mathbf{e}_{r_i} \mathbf{1}_{1 \times T_{ir_i}})^t$, in which $\mathbf{e}_k$ is a column vector of all 0's, except for a 1 in the k th position, and $\mathbf{1}_{r \times s}$ is an $r \times s$ matrix of all 1's. When it is clear by the context, we write $\mathbf{1}$ for $\mathbf{1}_{r \times s}$. With this, define $\mathbf{Z}_i = (\mathbf{Z}_{i1} \mathbf{0})$, in which $\mathbf{0}$ is a $T_i \times (q - r_i)$ zero matrix.

As in Section 2.2.2, let $\mathbf{R}_i = \text{Var}(\boldsymbol{\varepsilon}_i) = \sigma_\varepsilon^2 \text{diagonal}(1/w_{ijt})$. Write $\text{Var}(\mu_i) = \sigma_\mu^2$, and $\text{Var}(\gamma_{ij}) = \sigma_\gamma^2$. Then, $\mathbf{D} = \text{Var}(\alpha_i)$ is $\mathbf{D} = \sigma_\mu^2 \mathbf{1}_{q \times q} + \sigma_\gamma^2 \mathbf{I}_q$. One obtains that b_{GLS} is a weighted average of the y_{ijt} 's; see Table 1.

3. Prediction

To further establish the connection to credibility theory, we review the general prediction problem and its application to longitudinal data models. Then, we apply the results to the credibility models considered in Section 2.2.

3.1. General linear prediction problem

We develop the predictors by examining the minimum mean square error predictor of a generic random variable, W . The development we give is based on Harville's discussion of Robinson (1991). The argument is originally due to Goldberger (1962), who coined the phrase best linear unbiased predictor. The acronym BLUP was first used by Henderson (1973).

Consider a general situation in which we observe an $N \times 1$ random vector $\mathbf{y}$ with mean $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ and variance–covariance matrix $\text{Var}(\mathbf{y}) = \mathbf{V}$. We wish to predict a random variable W , that has mean $E(W) = \boldsymbol{\lambda}^t \boldsymbol{\beta}$ and $\text{Var}(W) = \sigma_W^2$. Denote the vector of covariances between W and $\mathbf{y}$ as $\text{Cov}(W, \mathbf{y}) = \text{Cov}_{W\mathbf{y}}$. The best linear (in $\mathbf{y}$) predictor of W is

$$W^* = E(W) + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} [\mathbf{y} - E(\mathbf{y})] = \boldsymbol{\lambda}^t \boldsymbol{\beta} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

If $(W, \mathbf{y}^t)^t$ is multivariate normal, then W^* equals $E(W|\mathbf{y})$ and, hence, is a minimum mean square predictor of W . The predictor W^* is also a minimum mean square predictor of W without the assumption of normality; see, for example, Norberg (1980) (Lemma 3.1). In the credibility literature, this formula for W^* is well known and is called the inhomogeneous estimator of W ; see, for example, Dannenburg et al. (1996).

To develop the BLUP, the generalized least squares estimator of $\boldsymbol{\beta}$ is $\mathbf{b}_{\text{GLS}} = (\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} (\mathbf{X}^t \mathbf{V}^{-1} \mathbf{y})$. This is the best linear unbiased estimator of $\boldsymbol{\beta}$. Replace $\boldsymbol{\beta}$ by $\mathbf{b}_{\text{GLS}}$ in the definition of W^* to get the BLUP of W ,

$$W_{\text{BLUP}} = \boldsymbol{\lambda}^t \mathbf{b}_{\text{GLS}} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X} \mathbf{b}_{\text{GLS}}) = (\boldsymbol{\lambda}^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \mathbf{b}_{\text{GLS}} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y}. \quad (3)$$

See Appendix A for a proof that W_{BLUP} is the best linear unbiased predictor of W . In the credibility literature, W_{BLUP} is called the homogeneous estimator of W (Dannenburg et al., 1996). From Appendix B, we also have the

form for the minimum mean square error

$$\text{Var}(W_{\text{BLUP}} - W) = (\boldsymbol{\lambda}^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X})(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} (\boldsymbol{\lambda}^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X})^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \text{Cov}_{W\mathbf{y}}^t + \sigma_W^2. \quad (4)$$

3.2. Longitudinal data prediction

To see how these general results apply to the longitudinal data model, we first introduce the mixed linear model. A more concise (stacked) matrix form of Eq. (2) is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\varepsilon},$$

in which $\mathbf{y} = (\mathbf{y}_1^t, \dots, \mathbf{y}_n^t)^t$ is the column vector of observations. Similarly, $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}_1^t, \dots, \boldsymbol{\varepsilon}_n^t)^t$. Here, $\mathbf{X}$ is the stacked matrix, $\mathbf{X} = (\mathbf{X}_1^t, \dots, \mathbf{X}_n^t)^t$, $\mathbf{Z}$ is the block diagonal matrix, $\mathbf{Z} = \text{block diagonal}(\mathbf{Z}_1, \dots, \mathbf{Z}_n)$, and $\boldsymbol{\alpha}$ is the stacked matrix $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_1^t, \dots, \boldsymbol{\alpha}_n^t)^t$. With this notation, $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Var}(\mathbf{y}) = \mathbf{V}$. The variance–covariance matrix is block diagonal of the form $\mathbf{V} = \text{block diagonal}(\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_n)$, where $\mathbf{V}_i = \mathbf{Z}_i \mathbf{D} \mathbf{Z}_i^t + \mathbf{R}_i$.

Suppose that we wish to predict the subject-specific effect $\boldsymbol{\alpha}_i$. Then, we consider linear combinations of the form $W = \mathbf{c}^t \boldsymbol{\alpha}_i$. Straightforward calculations show that $E(W) = 0$ so that $\boldsymbol{\lambda} = \mathbf{0}$. Further, $\text{Cov}(W, \mathbf{y}_j) = \mathbf{c}^t \mathbf{D} \mathbf{Z}_i^t$ for $j = i$, and $\text{Cov}(W, \mathbf{y}_j) = \mathbf{0}$, for $j \neq i$. Thus,

$$\text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y} = \mathbf{c}^t \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{y}_i \quad \text{and} \quad \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X} = \mathbf{c}^t \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i.$$

It follows that

$$W_{\text{BLUP}} = \mathbf{c}^t \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \mathbf{b}_{\text{GLS}}).$$

Because this is true for all vectors of constants $\mathbf{c}^t$, the BLUP of $\boldsymbol{\alpha}_i$ is the vector

$$\mathbf{a}_{i,\text{BLUP}} = \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \mathbf{b}_{\text{GLS}}).$$

By straightforward substitution, the associated mean square error is

$$\text{Var}(\mathbf{a}_{i,\text{BLUP}} - \boldsymbol{\alpha}_i) = \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i (\mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i)^{-1} \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{Z}_i \mathbf{D} - \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{Z}_i \mathbf{D} + \mathbf{D}.$$

Example 3 (Balanced Bühlmann, continued). In the balanced Bühlmann case, we have

$$\mathbf{a}_{i,\text{BLUP}} = \zeta (\bar{\mathbf{y}}_i - \bar{\mathbf{y}}),$$

in which $\zeta = T/(T + \sigma_\varepsilon^2/\sigma_\alpha^2)$ and $\bar{\mathbf{y}}_i = (1/T) \sum_{t=1}^T \mathbf{y}_{it}$ the sample average of subject i 's claims.

3.3. Credibility ratemaking application

In credibility ratemaking, one is interested in predicting the expected claims, conditional on the random risk parameters of the i th subject, for time period $T_i + 1$, say. In the longitudinal data context, then, the credibility ratemaking problem is to estimate $E(\mathbf{y}_{i,T_i+1} | \boldsymbol{\alpha}_i) = \mathbf{x}_{i,T_i+1} \boldsymbol{\beta} + \mathbf{z}_{i,T_i+1} \boldsymbol{\alpha}_i = W$. Thus, with this choice of W , we have $\boldsymbol{\lambda}^t = \mathbf{x}_{i,T_i+1}$. This yields $W_{\text{BLUP}} = \mathbf{x}_{i,T_i+1} \mathbf{b}_{\text{GLS}} + \mathbf{z}_{i,T_i+1} \mathbf{a}_{i,\text{BLUP}}$.

Example 4 (Balanced Bühlmann, continued). In the balanced Bühlmann case, we have $x_{it} = z_{it} = 1$, so that

$$W_{\text{BLUP}} = (1 - \zeta) \bar{\mathbf{y}} + \zeta \bar{\mathbf{y}}_i.$$

Here, recall that $\zeta = T/(T + \sigma_\varepsilon^2/\sigma_\alpha^2)$ is the credibility factor.

Table 2

Best linear unbiased predictors (BLUPs) and prediction of claims

Model	Notation	BLUPs	Prediction of claims
Bühlmann	$\zeta_i, \bar{y}_i, \bar{y}_w$ are as in Table 1	$a_{i, \text{BLUP}} = \zeta_i (\bar{y}_i - \bar{y}_w)$	For subject i , $(1 - \zeta_i) \bar{y}_w + \zeta_i \bar{y}_i$
Bühlmann–Straub	$\zeta_i, \bar{y}_{i,w}, \bar{y}_w$ are as in Table 1	$a_{i, \text{BLUP}} = \zeta_i (\bar{y}_{i,w} - \bar{y}_w)$	For subject i , $(1 - \zeta_i) \bar{y}_w + \zeta_i \bar{y}_{i,w}$
Hachemeister (Linear trend)	$\zeta_i, \mathbf{W}_i, \mathbf{b}_{\text{GLS}}$ are as in Table 1	$a_{i, \text{BLUP}} = \zeta_i \left(\mathbf{W}_i^{-1} \left[\sum_t w_{it} y_{it} \right] - \mathbf{b}_{\text{GLS}} \right)$	For period $T_i + 1, [1 T_i + 1](\mathbf{b}_{\text{GLS}} + a_{i, \text{BLUP}})$ where $\mathbf{b}_{\text{GLS}} + a_{i, \text{BLUP}}$ $= (1 - \zeta_i) \mathbf{b}_{\text{GLS}} + \zeta_i \mathbf{W}_i^{-1} \left[\sum_{t=1}^{T_i} w_{it} y_{it} \right]$
Jewell	$\zeta_i, \zeta_{ij}, \bar{y}_{i,w}, \bar{y}_{ij,w}, \bar{y}_w$ are as in Table 1	BLUP for $\mu_i, m_{i, \text{BLUP}} = \zeta_i (\bar{y}_{i,w} - \bar{y}_w)$ BLUP for $\gamma_{ij},$ $g_{ij, \text{BLUP}} = \zeta_{ij} (\bar{y}_{ij,w} - \zeta_i \bar{y}_{i,w} - (1 - \zeta_i) \bar{y}_w)$	For sector i , $b_{\text{GLS}} + m_{i, \text{BLUP}} = (1 - \zeta_i) \bar{y}_w + \zeta_i \bar{y}_{i,w}$ For subject j in sector i , $b_{\text{GLS}} + m_{i, \text{BLUP}} + g_{ij, \text{BLUP}}$ $= (1 - \zeta_i)(1 - \zeta_{ij}) \bar{y}_w + \zeta_i (1 - \zeta_{ij}) \bar{y}_{i,w}$ $+ \zeta_{ij} \bar{y}_{ij,w}$

Example 5 (Hachemeister, continued). In the Hachemeister case, we have $\mathbf{x}_{it} = \mathbf{z}_{it}$. Define $\mathbf{b}_i = (\mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i)^{-1} \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{y}_i$ to be the GLS estimator of $\boldsymbol{\alpha}_i + \boldsymbol{\beta}$ based only on the i th subject. The credibility estimator of $\boldsymbol{\alpha}_i + \boldsymbol{\beta}$ is

$$\begin{aligned} a_{i, \text{BLUP}} + \mathbf{b}_{\text{GLS}} &= \mathbf{D} \mathbf{X}_i^t \mathbf{V}_i^{-1} (\mathbf{y}_i - \mathbf{X}_i \mathbf{b}_{\text{GLS}}) + \mathbf{b}_{\text{GLS}} \\ &= \mathbf{b}_{\text{GLS}} - \mathbf{D} \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i \mathbf{b}_{\text{GLS}} + \mathbf{D} \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i \mathbf{b}_i = (\mathbf{I} - \zeta_i) \mathbf{b}_{\text{GLS}} + \zeta_i \mathbf{b}_i, \end{aligned}$$

in which $\zeta_i = \mathbf{D} \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i$ is the credibility factor. As in the Bühlmann case, again we see that the credibility estimator is a weighted average of a subject-specific statistic and a statistic that summarizes information from all subjects. This example is prominent in credibility theory because one can further express the GLS estimator of $\boldsymbol{\beta}$ as a weighted average of the $\mathbf{b}_i$ using the credibility factors as weights

$$\mathbf{b}_{\text{GLS}} = \left(\sum_{i=1}^n \zeta_i \right)^{-1} \sum_{i=1}^n \zeta_i \mathbf{b}_i.$$

See Taylor (1977) and Norberg (1980) for early discussions of this in the actuarial literature.

In Table 2, we show how to predict expected claims in the other examples that we considered in Section 2.2. Note that in each case, the predicted claims for the i th subject is a weighted average of that subject's experience with the $\mathbf{b}_{\text{GLS}}$, using the i th credibility factor as a weight.

From Eq. (4), with straightforward substitution, we also have the form for the minimum mean square error:

$$\begin{aligned} \text{Var}(W_{\text{BLUP}} - W) &= (\mathbf{x}_{i, T_i+1} - \mathbf{z}_{i, T_i+1} \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i) \left(\sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i \right)^{-1} \\ &\quad \times (\mathbf{x}_{i, T_i+1} - \mathbf{z}_{i, T_i+1} \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{X}_i)^t - \mathbf{z}_{i, T_i+1} \mathbf{D} \mathbf{Z}_i^t \mathbf{V}_i^{-1} \mathbf{Z}_i \mathbf{D} \mathbf{z}_{i, T_i+1}^t + \mathbf{z}_{i, T_i+1} \mathbf{D} \mathbf{z}_{i, T_i+1}^t. \quad (5) \end{aligned}$$

4. Estimation of variance components

In this section, we describe two methods for estimating the variance components. Both methods entail maximizing a likelihood function, in contrast to moment estimators that are commonly used in actuarial science; see, for example, Dannenburg (1994). In statistical estimation theory (Lehmann, 1991), there are well-known trade-offs

when considering moment compared to likelihood estimation. This section supplements moment estimators by introducing likelihood estimators in the actuarial literature. The likelihood function is maximized by using an iterative procedure that requires starting values. At the end of this section, we describe how to obtain reasonable starting values for the iteration.

4.1. Maximum likelihood estimation assuming normality

Assuming joint normality of the subject-specific effects and the errors, the log-likelihood of a single subject is

$$l_i(\boldsymbol{\beta}, \boldsymbol{\tau}) = -\frac{1}{2} [\ln \det(\mathbf{V}_i(\boldsymbol{\tau})) + (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})^t \mathbf{V}_i(\boldsymbol{\tau})^{-1} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta})],$$

up to an additive constant. Here, $\det$ is the determinant. Thus, the log-likelihood for the entire data set is $L(\boldsymbol{\beta}, \boldsymbol{\tau}) = \sum_{i=1}^n l_i(\boldsymbol{\beta}, \boldsymbol{\tau})$. The values of $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$ that maximize $L(\boldsymbol{\beta}, \boldsymbol{\tau})$ are the maximum likelihood estimators (MLEs).

The “score” vector is the vector of derivatives with respect to the parameters. We denote the vector of parameters by $\boldsymbol{\theta} = (\boldsymbol{\beta}^t, \boldsymbol{\tau}^t)^t$. With this notation, the score vector is $\partial L(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$. If this score has a root, then the root is a maximum likelihood estimator. To compute the score vector, we first take derivatives with respect to $\boldsymbol{\beta}$ and find the root. That is,

$$\frac{\partial L(\boldsymbol{\beta}, \boldsymbol{\tau})}{\partial \boldsymbol{\beta}} = \frac{\partial}{\partial \boldsymbol{\beta}} \sum_{i=1}^n l_i(\boldsymbol{\beta}, \boldsymbol{\tau}) = \sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i(\boldsymbol{\tau})^{-1} (\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) = 0.$$

This yields $\mathbf{b}_{\text{MLE}}(\boldsymbol{\tau}) = (\sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i(\boldsymbol{\tau})^{-1} \mathbf{X}_i)^{-1} (\sum_{i=1}^n \mathbf{X}_i^t \mathbf{V}_i(\boldsymbol{\tau})^{-1} \mathbf{y}_i) = \mathbf{b}_{\text{GLS}}(\boldsymbol{\tau})$. That is, for fixed variance components $\boldsymbol{\tau}$, the maximum likelihood estimators and the general least squares estimators are the same. This yields the so-called “concentrated” or “profile” log-likelihood

$$L(\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}), \boldsymbol{\tau}) = -\frac{1}{2} \sum_{i=1}^n [\ln \det(\mathbf{V}_i(\boldsymbol{\tau})) + (\text{Error SS})_i(\boldsymbol{\tau})],$$

a function of $\boldsymbol{\tau}$. Here, the error sum of squares is

$$(\text{Error SS})_i(\boldsymbol{\tau}) = (\mathbf{y}_i - \mathbf{X}_i \mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}))^t \mathbf{V}_i(\boldsymbol{\tau})^{-1} (\mathbf{y}_i - \mathbf{X}_i \mathbf{b}_{\text{GLS}}(\boldsymbol{\tau})).$$

Thus, we now maximize the log-likelihood as a function of $\boldsymbol{\tau}$ only. In some cases, one can obtain closed form expressions for the maximizing variance components, as shown in the following example.

Example 6 (Balanced Bühlmann, continued). In the balanced Bühlmann case, the MLE of the error variance is

$$(\hat{\sigma}_\varepsilon^2)_{\text{MLE}} = \frac{\sum_{i=1}^n (\sum_{t=1}^T (y_{it} - \bar{y}_i)^2)}{n(T-1)},$$

a familiar unbiased estimator of the process variance (Klugman et al., 1998). However, the MLE of the variance of the hypothetical means is biased

$$(\hat{\sigma}_\alpha^2)_{\text{MLE}} = \frac{\sum_{i=1}^n (\bar{y}_i - \bar{y})^2}{n} - \frac{\sum_{i=1}^n (\sum_{t=1}^T (y_{it} - \bar{y}_i)^2)}{nT(T-1)}.$$

In general, the variance components are estimated recursively. This can be done using either the Newton–Raphson or Fisher scoring method (Harville, 1977; Wolfinger et al., 1994).

Newton–Raphson: Let $L = L(\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}), \boldsymbol{\tau})$, and use the iterative method:

$$\boldsymbol{\tau}_{\text{NEW}} = \boldsymbol{\tau}_{\text{OLD}} - \left\{ \left(\frac{\partial^2 L}{\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}^t} \right)^{-1} \frac{\partial L}{\partial \boldsymbol{\tau}} \right\} \bigg|_{\boldsymbol{\tau}=\boldsymbol{\tau}_{\text{OLD}}}$$

Here, the matrix $\partial^2 L / (\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}')$ is called the “sample information matrix.”

Fisher scoring: Define the expected information matrix $H(\boldsymbol{\tau}) = E(\partial^2 L / (\partial \boldsymbol{\tau} \partial \boldsymbol{\tau}'))$, and use

$$\boldsymbol{\tau}_{\text{NEW}} = \boldsymbol{\tau}_{\text{OLD}} - H(\boldsymbol{\tau}_{\text{OLD}})^{-1} \left\{ \frac{\partial L}{\partial \boldsymbol{\tau}} \right\} \bigg|_{\boldsymbol{\tau}=\boldsymbol{\tau}_{\text{OLD}}}.$$

In the next section, we learn how to obtain unbiased estimators of the variance components.

4.2. Restricted maximum likelihood

As we saw in Section 3, credibility factors are based on variance components $\boldsymbol{\tau}$. Thus, estimation of these variance components is critical to credibility theory. Several estimation techniques have been developed in the statistical and actuarial literatures, including Swamy’s (1970) moment estimators, Rao’s (1970) minimum norm, quadratic unbiased estimators, DeVyllder’s (1981) pseudo-estimators, and Norberg’s (1980, 1982) estimators. In this section, we discuss a method that is widely employed in the statistical literature, although not cited in the actuarial literature.

As the name suggests, *restricted maximum likelihood* (REML) is a likelihood-based estimation procedure. Thus, it shares many of the desirable properties of maximum likelihood estimators. Because it is based on likelihoods, it is not specific to a particular design matrix, as are the analysis of variance estimators (Harville, 1977). Thus, it can be readily applied to a wide variety of models. Like MLEs, REML estimators are translation invariant.

Maximum likelihood often produces biased estimators of the variance components, $\boldsymbol{\tau}$, as we saw in the balanced Bühlmann case. In contrast, estimation based on REML results in unbiased estimators of $\boldsymbol{\tau}$, at least for many balanced designs. Because maximum likelihood estimators are negatively biased, they often turn out to be negative, an intuitively undesirable situation for many users. Because of the unbiasedness of many REML estimators, there is less of a tendency to produce negative estimators (Corbeil and Searle, 1976a). As with MLEs, REML estimators can be defined to be parameter values for which the (restricted) likelihood achieves a maximum value over a constrained parameter space. Thus, as with maximum likelihood, it is straightforward to modify the method to produce nonnegative variance estimators.

The idea behind REML estimation is to consider the likelihood of linear combinations of the responses that do not depend on the mean parameters. To illustrate, consider the following generic situation: The responses, denoted by the vector $\mathbf{y}$, are normally distributed, have mean $E(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}$ and variance–covariance matrix $\text{Var}(\mathbf{y}) = \mathbf{V} = \mathbf{V}(\boldsymbol{\tau})$. The dimension of $\mathbf{y}$ is $N \times 1$, and the dimension of $\mathbf{X}$ is $N \times p$. With this notation, define the projection matrix $\mathbf{Q} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ and consider the linear combination of responses $\mathbf{Q}\mathbf{y}$. Some straightforward calculations show that this has mean $\mathbf{0}$ and variance–covariance matrix $\text{Var}(\mathbf{Q}\mathbf{y}) = \mathbf{Q}\mathbf{V}\mathbf{Q}$. Because (i) $\mathbf{Q}\mathbf{y}$ has a multivariate normal distribution and (ii) the mean and variance–covariance matrix do not depend on $\boldsymbol{\beta}$, the distribution of $\mathbf{Q}\mathbf{y}$ does not depend on $\boldsymbol{\beta}$. Further, it turns out that $\mathbf{Q}\mathbf{y}$ is independent of the generalized least squares estimator $\mathbf{b}_{\text{GLS}} = (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y}$ (Appendix C).

The vector $\mathbf{Q}\mathbf{y}$ is the residual vector in an ordinary least squares fit of the data. Hence, REML is also referred to as *residual maximum likelihood estimation*. Because the rank of $\mathbf{Q}$ is $N - p$, we lose some information by considering this transformation of the data. Hence, the term *restricted maximum likelihood*. (There is some information about $\boldsymbol{\tau}$ in the vector $\mathbf{b}_{\text{GLS}}$ that we are not using for estimation.) Further, note that we could also use any linear transform of $\mathbf{Q}$, such as $\mathbf{A}\mathbf{Q}$, in that $\mathbf{A}\mathbf{Q}\mathbf{y}$ also has a multivariate normal distribution with a mean and variance–covariance matrix that do not depend on $\boldsymbol{\beta}$. Patterson and Thompson (1971) and Harville (1974, 1977) showed that the likelihood does not depend on the choice of $\mathbf{A}$. They introduced the “restricted” log-likelihood

$$L_{\text{REML}}(\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}), \boldsymbol{\tau}) = \frac{1}{2} [\ln \det(\mathbf{V}(\boldsymbol{\tau})) + \ln \det(\mathbf{X}'\mathbf{V}(\boldsymbol{\tau})^{-1}\mathbf{X}) + (\text{Error SS})(\boldsymbol{\tau})],$$

up to an additive constant. See Appendix D for a derivation of this likelihood. REML estimators $\boldsymbol{\tau}_{\text{REML}}$ are defined to be maximizers of the function $L_{\text{REML}}(\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}), \boldsymbol{\tau})$. Here, the error sum of squares is

$$(\text{Error SS})(\boldsymbol{\tau}) = (\mathbf{y} - \mathbf{X}\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}))' \mathbf{V}(\boldsymbol{\tau})^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau})).$$

For comparison, the usual log-likelihood is

$$L(\mathbf{b}_{\text{GLS}}(\boldsymbol{\tau}), \boldsymbol{\tau}) = -\frac{1}{2} [\ln \det(\mathbf{V}(\boldsymbol{\tau})) + (\text{Error SS})(\boldsymbol{\tau})],$$

up to an additive constant. The only difference between the two likelihoods is the term $\ln \det(\mathbf{X}'\mathbf{V}(\boldsymbol{\tau})^{-1}\mathbf{X})$; thus, iterative methods of maximization are the same – that is, using Newton–Raphson or Fisher scoring. For longitudinal data models, this additional term is $\sum_{i=1}^n \ln \det(\mathbf{X}_i' \mathbf{V}_i(\boldsymbol{\tau})^{-1} \mathbf{X}_i)$.

For balanced analysis of variance data ($T_i = T$), Corbeil and Searle (1976a) established that the REML estimation reduces to standard analysis of variance estimators. Thus, REML estimators are unbiased for these designs. However, REML estimators and analysis of variance estimators differ for unbalanced data. REML estimators achieve their unbiasedness by accounting for the degrees of freedom lost in estimating the fixed effects $\boldsymbol{\beta}$; MLEs do not account for this loss of degrees of freedom. When p is large, the difference between REML estimators and MLEs is significant. Corbeil and Searle (1976b) showed that, in terms of mean square errors, MLEs outperform REML estimators for small p (< 5), although the situation is reversed for large p with a sufficiently large sample.

Harville (1974) gave a Bayesian interpretation of REML estimators. He pointed out that using only $\mathbf{Qy}$ to make inferences about $\boldsymbol{\tau}$ is equivalent to ignoring prior information about $\boldsymbol{\beta}$ and using all the data.

Example 7 (Balanced Bühlmann, continued). In the balanced Bühlmann case, the REML estimator of the error variance equals the MLE

$$(\hat{\sigma}_\varepsilon^2)_{\text{REML}} = \frac{\sum_{i=1}^n (\sum_{t=1}^T (y_{it} - \bar{y}_i)^2)}{n(T-1)} = (\hat{\sigma}_\varepsilon^2)_{\text{MLE}}.$$

The REML estimator of the variance of the hypothetical means is unbiased, as expected

$$(\hat{\sigma}_\alpha^2)_{\text{REML}} = \frac{\sum_{i=1}^n (\bar{y}_i - \bar{y})^2}{n-1} - \frac{\sum_{i=1}^n (\sum_{t=1}^T (y_{it} - \bar{y}_i)^2)}{nT(T-1)}.$$

These estimators are the ones usually applied in credibility theory (Bühlmann and Straub, 1970).

4.3. Starting values

Both the Newton–Raphson and Fisher scoring algorithms and the ML and REML estimation methods involve recursive calculations that require starting values. In this section, we describe two non-recursive methods due to Swamy (1970) and Rao (1970). One can use the results of these non-recursive methods as starting values in the Newton–Raphson and Fisher scoring algorithms.

Swamy's moment-based procedure appeared in the econometrics panel data literature. Related methods have also appeared in the actuarial literature; see, for example, Goovaerts and Hoogstad (1987) or Dannenburg et al. (1996) for general references. We consider a random coefficients model; that is, Eq. (2) with $\mathbf{x}_{it} = \mathbf{z}_{it}$ and $\mathbf{R}_i = \sigma_i^2 \mathbf{I}_{T_i}$. For this procedure, one performs the following steps:

- (i) Use an ordinary least squares estimator of σ_i^2 ,

$$s_i^2 = \frac{\mathbf{y}_i' (\mathbf{I}_{T_i} - \mathbf{X}_i (\mathbf{X}_i' \mathbf{X}_i)^{-1} \mathbf{X}_i') \mathbf{y}_i}{T_i - K}.$$

(This is an ordinary least squares procedure in that it ignores $\mathbf{D}$.)

- (ii) To estimate $\mathbf{D}$, we first calculate $\mathbf{b}_{i,\text{OLS}} = (\mathbf{X}_i' \mathbf{X}_i)^{-1} \mathbf{X}_i' \mathbf{y}_i$, a predictor of $\boldsymbol{\beta} + \boldsymbol{\alpha}_i$.
- (iii) Next, we estimate $\mathbf{D}$ using $\mathbf{D}_{\text{SWAMY}} = \frac{\sum_{i=1}^n (\mathbf{b}_{i,\text{OLS}} - \bar{\mathbf{b}})(\mathbf{b}_{i,\text{OLS}} - \bar{\mathbf{b}})'}{n-1} - (1/n) \sum_{i=1}^n s_i^2 (\mathbf{X}_i' \mathbf{X}_i)^{-1}$, where $\bar{\mathbf{b}} = (1/n) \sum_{i=1}^n \mathbf{b}_{i,\text{OLS}}$.

The estimator of $\mathbf{D}$ can be motivated by examining the variance of $\mathbf{b}_{i,\text{OLS}}$,

$$\text{Var}(\mathbf{b}_{i,\text{OLS}}) = \text{Var}((\mathbf{X}_i^t \mathbf{X}_i)^{-1} \mathbf{X}_i^t (\boldsymbol{\beta} + \boldsymbol{\alpha}_i) + \boldsymbol{\varepsilon}_i) = \text{Var}(\boldsymbol{\beta} + \boldsymbol{\alpha}_i + (\mathbf{X}_i^t \mathbf{X}_i)^{-1} \mathbf{X}_i^t \boldsymbol{\varepsilon}_i) = \mathbf{D} + \sigma_i^2 (\mathbf{X}_i^t \mathbf{X}_i)^{-1}.$$

Using $\frac{\sum_{i=1}^n (\mathbf{b}_{i,\text{OLS}} - \bar{\mathbf{b}})(\mathbf{b}_{i,\text{OLS}} - \bar{\mathbf{b}})^t}{n-1}$ and s_i^2 as estimators of $\text{Var}(\mathbf{b}_{i,\text{OLS}})$ and σ_i^2 , respectively, yields $\mathbf{D}_{\text{SWAMY}}$ as an estimator of $\mathbf{D}$.

Various modifications of this estimator are possible. One can iterate the procedure by using $\mathbf{D}_{\text{SWAMY}}$ to improve the estimators s_i^2 and so on. Homoscedasticity of the ε_i terms could also be assumed. Hsiao (1986) recommends dropping the second term, $(1/n) \sum_{i=1}^n s_i^2 (\mathbf{X}_i^t \mathbf{X}_i)^{-1}$, to ensure that $\mathbf{D}_{\text{SWAMY}}$ is non-negative definite.

Another non-recursive method is Rao's (1970) *minimum variance quadratic unbiased estimator* (MIVQUE). To describe this method, we return to the general linear model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, in which $\text{Var}(\mathbf{y}) = \mathbf{V} = \mathbf{V}(\boldsymbol{\tau})$. We wish to estimate the linear combination of variance components, $\sum_{k=1}^r c_k \tau_k$, where the c_k are specified constants and $\boldsymbol{\tau} = (\tau_1, \dots, \tau_r)^t$. We assume that $\mathbf{V}$ is linear in the sense that

$$\mathbf{V} = \sum_{k=1}^r \tau_k \frac{\partial}{\partial \tau_k} \mathbf{V}.$$

Thus, with this assumption, we have that the Hessian of $\mathbf{V}$ is zero (Graybill, 1969). Although this assumption works often, it is not satisfied by, for example, autoregressive models. It is not restrictive to assume that $(\partial/\partial \tau_k) \mathbf{V}$ is known even though the variance component τ_k is unknown.

Quadratic estimators of $\sum_{k=1}^r c_k \tau_k$ are based on $\mathbf{y}' \mathbf{A} \mathbf{y}$, where $\mathbf{A}$ is a symmetric matrix to be specified. The variance of $\mathbf{y}' \mathbf{A} \mathbf{y}$, assuming normality, can easily be shown to be $2 \text{trace}(\mathbf{V} \mathbf{A} \mathbf{V} \mathbf{A})$. We would like the estimator to be invariant to translation of $\boldsymbol{\beta}$. That is, we require that $\mathbf{y}' \mathbf{A} \mathbf{y} = (\mathbf{y} - \mathbf{X} \mathbf{b}_0)' \mathbf{A} (\mathbf{y} - \mathbf{X} \mathbf{b}_0)$ for each $\mathbf{b}_0$. Thus, we restrict our choice of $\mathbf{A}$ to those that satisfy $\mathbf{A} \mathbf{X} = \mathbf{0}$.

For unbiasedness, we would like $\sum_{k=1}^r c_k \tau_k = E(\mathbf{y}' \mathbf{A} \mathbf{y})$. Using $\mathbf{A} \mathbf{X} = \mathbf{0}$, we have

$$E(\mathbf{y}' \mathbf{A} \mathbf{y}) = E(\boldsymbol{\varepsilon}' \mathbf{A} \boldsymbol{\varepsilon}) = \text{trace}(E(\boldsymbol{\varepsilon} \boldsymbol{\varepsilon}' \mathbf{A})) = \text{trace}(\mathbf{V} \mathbf{A}) = \sum_{k=1}^r \tau_k \text{trace} \left(\left(\frac{\partial}{\partial \tau_k} \mathbf{V} \right) \mathbf{A} \right).$$

Because this equality should be valid for all variance components τ_k , we require $\mathbf{A}$ to satisfy

$$c_k = \text{trace} \left(\left(\frac{\partial}{\partial \tau_k} \mathbf{V} \right) \mathbf{A} \right) \quad \text{for } k = 1, \dots, r. \quad (6)$$

Rao showed that the minimum value of $\text{trace}(\mathbf{V} \mathbf{A} \mathbf{V} \mathbf{A})$ satisfying $\mathbf{A} \mathbf{X} = \mathbf{0}$ and the constraints in Eq. (6) is attained at

$$\mathbf{A}_*(\mathbf{V}) = \sum_{k=1}^r \lambda_k \mathbf{V}^{-1} \boldsymbol{\mathcal{Q}} \left(\frac{\partial}{\partial \tau_k} \mathbf{V} \right) \mathbf{V}^{-1} \boldsymbol{\mathcal{Q}},$$

where $\boldsymbol{\mathcal{Q}} = \boldsymbol{\mathcal{Q}}(\mathbf{V}) = \mathbf{I} - \mathbf{X}(\mathbf{X}' \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{V}^{-1}$ and $(\lambda_1, \dots, \lambda_r)$ is the solution of

$$\mathbf{S}(\lambda_1, \dots, \lambda_r)^t = (c_1, \dots, c_r)^t.$$

Here, the (i, j) th element of $\mathbf{S}$ is given by

$$\text{trace} \left(\mathbf{V}^{-1} \boldsymbol{\mathcal{Q}} \left(\frac{\partial}{\partial \tau_i} \mathbf{V} \right) \mathbf{V}^{-1} \boldsymbol{\mathcal{Q}} \left(\frac{\partial}{\partial \tau_j} \mathbf{V} \right) \right).$$

Thus, the MIVQUE estimator of $\boldsymbol{\tau}$ is the solution of

$$\mathbf{S} \boldsymbol{\tau}_{\text{MIVQUE}} = \mathbf{G}, \quad (7)$$

where the k th element of $\mathbf{G}$ is given by $\mathbf{y}' \mathbf{V}^{-1} \boldsymbol{\mathcal{Q}} ((\partial/\partial \tau_k) \mathbf{V}) \mathbf{V}^{-1} \boldsymbol{\mathcal{Q}} \mathbf{y}$.

When comparing Rao's to Swamy's method, we note that the MIVQUE estimators are available for a larger class of models. To illustrate, in the longitudinal data context, it is possible to handle serial correlation with the MIVQUE estimators. A drawback of the MIVQUE estimator is that normality is assumed; this can be weakened to zero kurtosis for certain forms of $\mathbf{V}$ (Swallow and Searle, 1978). Further, MIVQUE estimators require a pre-specified estimate of $\mathbf{V}$. A widely used specification is to use the identity matrix for $\mathbf{V}$ in equation Eq. (7). This specification produces so-called MIVQUE(0) estimators, an option in widely available statistical packages.

5. Example: Hachemeister linear model

In this section, we implement the estimation procedures from Section 4 and show how to use the resulting estimates to produce the credibility-adjusted premiums from Section 3. For our analysis, we used the statistical package SAS; specifically, we used a procedure within SAS called PROC MIXED. For practicing actuaries, an important point of interpreting credibility models as longitudinal data models is that specialized software is not required; widely available statistical packages can be used to produce credibility estimators.

We consider the well-known data set of Hachemeister (1975) and apply his linear trend model to the data. This data set has been used in the actuarial literature to illustrate credibility techniques. In addition to Hachemeister (1975), Goovaerts and Hoogstad (1987) applied several credibility techniques to these data. Dannenburg et al. (1996) also analyzed these data when they presented Hachemeister's model.

Hachemeister analyzed claim data from private passenger auto insurance, bodily injury coverage. The data consists of information from $n = 5$ states in the US during $T = 12$ quarters, from the third quarter of 1970 through the second quarter of 1973, inclusive. The response of interest is the average loss per claim, or severity. Fig. 1 displays a multiple time series plot of the response variable for States 1 and 3. State 1 has the highest severity in most quarters. State 3 has larger variability in its claims than does State 1. Further, there is a small increase in severity over time that Hachemeister ascribed to inflation. Differences among states and increases over time suggested to Hachemeister the model

$$y_{it} = c_{i1} + c_{i2}t + \varepsilon_{it}, \quad (8)$$

where $i = 1, \dots, 5$, and $t = 1, \dots, 12$. This is our Eq. (1) with $K = q = 2$, $\mathbf{x}_{it} = \mathbf{z}_{it} = (1 \ t)$, and $\mathbf{c}_i = \boldsymbol{\beta} + \boldsymbol{\alpha}$.

Bühlmann and Gisler (1997) suggest (approximate) centering of the time variable. This results in near orthogonality between the explanatory vectors and a constant vector, yielding credibility factors that are more intuitively appealing to users. However, this intuitive appeal relies in part on the assumption of independence of variance parameters which seems inappropriate for the Hachemeister data. For this reason, and to enhance comparability with the Goovaerts and Hoogstad results, we did not use their proposed transformation.

We estimated this model using various combinations of

- (i) estimation methods – restricted maximum likelihood (REML) versus maximum likelihood (ML),
- (ii) iteration methods – Newton–Raphson (NR) versus Fisher scoring (FS), and
- (iii) starting values – Swamy's moment estimators versus Rao's minimum variance quadratic unbiased estimators, MIVQUE(0).

We also estimated the model by constraining the estimator of the variance–covariance matrix $\mathbf{D}$, $\hat{\mathbf{D}}$, to be positive definite. Specifically, we constrained $\hat{\mathbf{D}}$ by using the lower Cholesky decomposition of $\hat{\mathbf{D}}$, such that $\hat{\mathbf{D}} = \hat{\mathbf{L}}\hat{\mathbf{L}}^t$ (Graybill, 1969). We estimated the model by requiring or not requiring each diagonal term of $\hat{\mathbf{L}}$ to be positive. (The SAS defaults are REML, NR, MIVQUE(0) and no constraints on $\hat{\mathbf{L}}$.) The results are summarized in Table 3 in which we present the $\mathbf{b}_{\text{GLS}}$, the individual estimates for each state, and the credibility-adjusted estimates for each state. We list the estimate of the intercept first, then the slope. The base model given in Table 3 is the linear trend model estimated by Goovaerts and Hoogstad (1987).

For our study of the Hachemeister data, we found the REML and ML estimation methods gave similar results. We anticipated this because REML provides a correction for the degrees of freedom when estimating fixed regression

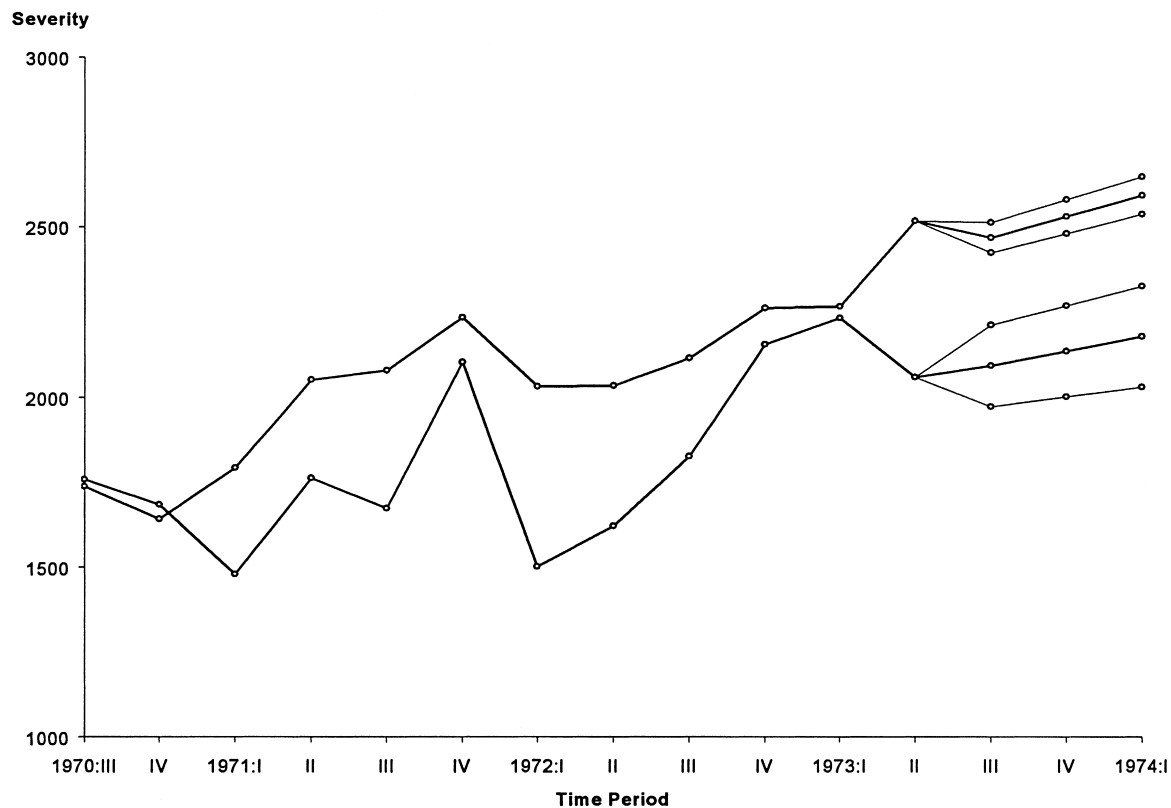


Fig. 1. Severity Predictions Based on Hachemeister's Model and Data. The thicker lines show Hachemeister data with one-, two-, and three-step predictions of severities for states 1 and 3. The thinner lines show the prediction intervals within one standard error of the prediction for each of these two states. The group of upper lines is for State 1. The group of lower lines is for State 3.

parameters, yet with this example, we had only $K = 2$ such parameters. Also not surprising was that Fisher scoring took longer to converge, generally in excess of 50 iterations of the recursive algorithm. It provided similar results to the Newton–Raphson routine.

Although SAS default MIVQUE(0) performed well in most trials, occasionally it produced non-feasible starting values, and thus the algorithm did not converge. For this random coefficient model, the moment estimator of Swamy provided a useful starting value. We used the modification suggested in Hsiao (1986), namely dropping the second term in the estimator of $\mathbf{D}$, to ensure our initial estimator was non-negative definite. For this data, efforts to include this term also precluded convergence of the algorithm. Thus, although the SAS default works for many applications, there are occasions where other starting value procedures are needed.

The most surprising occurrence of our estimation study was the effect of the constraints placed on $\hat{\mathbf{D}}$. When we did not constrain $\hat{\mathbf{D}}$ to be positive definite, the recursive algorithms estimated $\hat{\mathbf{D}}$ to be negative definite. In this case, our estimates are similar to other results published in the literature, such as Goovaerts and Hoogstad (1987) and Dannenburg et al. (1996). However, a negative definite matrix $\hat{\mathbf{D}}$ allows the estimated variance of linear combinations of α to be less than zero, an intuitively implausible situation for many practitioners. For a one-dimensional credibility factor, a negative variance results in a credibility factor outside the interval $[0, 1]$. Thus, it is customary in applications to constrain estimators of variance–covariance matrices such as $\hat{\mathbf{D}}$ to be positive definite. For this data, our resulting credibility-adjusted estimators and predictions are markedly different than without this constraint. Specifically, in Table 3, note that when the estimated matrix $\hat{\mathbf{D}}$ is not positive definite, we observe some strange behavior in the credibility-adjusted estimates for some of the states. In particular, for State 4, the credibility-adjusted estimate of

Table 3

Individual estimates, credibility estimates, and predictions of Hachemeister's data

Estimation ($\hat{D} > 0$) ^b b_{GLS}		Individual estimates						Credibility adjusted estimates					Prediction and standard errors				
procedure ^a		State 1	2	3	4	5	State 1	2	3	4	5	State 1	2	3	4	5	
Base	N	1469	1664	1400	1537	1174	1520	1696	1377	1540	1312	1421	2436	1650	2073	1507	1759
		32	62	17	43	28	12	57	21	41	15	26					
RNSN	N	1501	1658	1398	1533	1177	1522	1660	1434	1555	1394	1464	2464	1606	2067	1454	1720
		28	62	17	43	28	12	62	13	39	5	20	(41)	(76)	(92)	(156)	(59)
RNSY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1199	1518	2468	1624	2093	1528	1680
		32	62	17	43	28	121	62	18	43	25	12	(44)	(96)	(119)	(204)	(72)
RNMN	N	1501	1658	1398	1533	1177	1522	1660	1434	1555	1394	1464	2464	1606	2067	1454	1720
		28	62	17	43	28	12	62	13	39	5	20	(41)	(76)	(92)	(156)	(59)
RNMY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1199	1518	2468	1624	2093	1528	1680
		32	62	17	43	28	12	62	18	43	25	12	(44)	(96)	(119)	(204)	(72)
RFSN	N	1501	1658	1398	1533	1177	1522	1660	1434	1555	1394	1464	2464	1606	2067	1453	1720
		28	62	17	43	28	12	62	13	39	5	20	(41)	(76)	(92)	(157)	(59)
RFSY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1199	1518	2468	1624	2093	1527	1680
		32	62	17	43	28	12	62	18	43	25	12	(44)	(96)	(119)	(204)	(72)
RFMN	N	1579	1658	1398	1533	1177	1522	1579	1579	1579	1579	1579	2509	1534	2056	1395	1649
		19	62	17	43	28	12	72	−3	37	−14	5	(47)	(93)	(114)	(193)	(71)
RFMY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1199	1518	2468	1624	2093	1527	1680
		32	62	17	43	28	12	62	18	43	25	12	(44)	(96)	(119)	(204)	(72)
MNSN	N	1502	1658	1398	1533	1177	1522	1660	1435	1555	1397	1464	2463	1608	2065	1467	1720
		28	62	17	43	28	12	62	13	39	5	20	(41)	(75)	(91)	(153)	(59)
MNSY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1197	1518	2468	1624	2093	1530	1680
		32	62	17	43	28	12	62	18	43	26	12	(44)	(95)	(118)	(202)	(71)
MNMN	Yes, but Hessian is not positive definite – leads to trying to take the inverse a singular matrix																
MNMY	Initial estimate is not feasible																
MFSN	N	1502	1658	1398	1533	1177	1522	1660	1435	1555	1397	1464	2463	1608	2065	1467	1720
		28	62	17	43	28	12	62	13	39	5	20	(41)	(75)	(91)	(153)	(59)
MFSY	Y	1461	1658	1398	1533	1177	1522	1659	1396	1535	1197	1518	2468	1624	2093	1529	1680
		32	62	17	43	28	12	62	18	43	26	12	(44)	(95)	(118)	(201)	(71)
MFMN	N	1579	1658	1398	1533	1177	1522	1579	1579	1579	1579	1579	2508	1537	2053	1412	1650
		19	62	17	43	28	12	71	−3	36	−13	5	(46)	(91)	(112)	(188)	(70)
MFMY	Initial estimate is not feasible																

^a Estimation procedure=(Estimation Method)/(Iteration Method)/(Starting Value)/(Lower Bound); Estimation Method: R (REML, SAS default), M (ML), Iteration Method: N (Newton-Raphson, SAS default), F (Fisher scoring), Starting Value: S (Swamy), M (MIVQUE(0), SAS default), Lower Bound: N (No lower bound on L), Y (Yes, lower bound of 0.10 and 0.01 for L's diagonal elements).

^b $\hat{D} > 0$: Y means that the estimator of $\hat{D}$, $\hat{D}$, is constrained to be positive definite; N means that it is not constrained.

the slope is well below both the overall and individual estimates of the slope when the matrix $\hat{D}$ is not positive definite.

We also found that our estimates of the variance components were sensitive to the choice of constraints. When we set $\hat{D} = \hat{L}\hat{L}'$ with each diagonal term of $\hat{L}$ constrained to be positive, the variance component estimates were sensitive to the choice of these constraints. Further, it was interesting that REML estimation was more stable than ML estimation, in that the former converged in many instances when the latter did not.

Having interpreted the credibility ratemaking problem as a longitudinal data model, it is easy to generate many standard statistical measures that are not often displayed in the ratemaking context. To illustrate, in Fig. 1 we provide one-, two-, and three-step predictions of claims for States 1 and 3. Notice that the less variable state, State 1 with generally larger claims has smaller prediction intervals than State 3, which demonstrates more variability over time. Further, Table 3 provides standard errors for the one-step predictions. The standard errors are estimated values of the square roots of the minimum mean square errors given in Eq. (5). In Fig. 1, the prediction intervals are determined as BLUP's (credibility estimators) plus and minus one standard error. The purpose of the graphical display of the

prediction intervals is to provide users with a range of reliability of the predictors. This display helps actuaries communicate the uncertainty of their predictions to management.

6. Summary and conclusions

The purpose of this paper is to interpret credibility ratemaking models in the context of statistical longitudinal data models. The longitudinal data literature of greatest interest to actuaries is widely scattered. In this paper, we collected these results using common notation, and we related them to the relevant actuarial literature. Specifically, we also reviewed several credibility models and showed how to write these in the longitudinal data context. We reviewed the general prediction problem, showed how it applies to the longitudinal data setting, and derived classical credibility estimators within this framework.

A large portion of this paper is devoted to synthesizing results on parameter estimation of variance components because of its importance in applying longitudinal data models in practice. We reviewed iteration methods and provided a detailed summary of restricted maximum likelihood (REML), as well as techniques for starting the recursive maximization procedures. We illustrated the parameter estimation procedures by considering a well-known data set in the actuarial literature, Hachemeister's (1975) data. Based on our examination of this data, we recommend using the first two SAS defaults, REML estimation together with the Newton–Raphson iteration method. Having other starting values in addition to Rao's MIVQUE(0) is advisable and for our random coefficients model, we found Swamy's moment estimator to be useful. Depending on the application, actuaries may wish to constrain estimators of variance–covariance matrices to be positive definite. Standard statistical software allows this, although one needs to be careful with the choice of constraints.

The longitudinal data model is a special case of the general linear (mixed) model. Hence, it shares the advantages and disadvantages of this larger class of models. In particular, distributional assumptions on the responses are not required for the parameter estimators to be consistent and asymptotically normal, which are desirable large sample properties. For small samples, estimation can be dramatically improved by accounting for the distributional features of the data. To illustrate, for count data, estimators based on a Poisson regression model are generally preferable to those based on the linear model (McCullagh and Nelder, 1989). As another illustration, Kaas et al. (1997) considered models of exponential families for weighted responses although without regression covariates. For count and other types of data that can be modeled using exponential families, the generalized linear model (GLM) is a broad framework for accommodating a variety of distributional assumptions. Recently, Nelder and Verrall (1997) examined credibility estimators within the GLM framework. They described predictors based on the hierarchical GLM model of Lee and Nelder (1996); these predictors are best linear unbiased asymptotically, although not in finite samples. Further, the models they consider do not allow for serial correlation nor variable slopes. Of course, an important advantage of the GLM structure is that one can construct estimators that incorporate features of the data such as discreteness and the long-tailed nature of the distribution. Whether this framework can be used for prediction assuming compound distributions that are commonly encountered in actuarial practice is an open question.

There are other interesting links between credibility and longitudinal data models that we did not explore in this paper. One example consists of the time-varying parameter models that are useful for small cross-sections with a long history of information. These models are generally estimated using the Kalman filter. Ledolter et al. (1991) and Hsiao (1986) provide introductions to this approach from actuarial and econometric perspectives, respectively. Another example is the Bayesian interpretation of random coefficient models. See Klugman (1992) and Mátyás and Sevestre (1996) for further discussions from actuarial and econometric perspectives, respectively. To keep this paper focused, we examined only a frequentist perspective.

Longitudinal data models capture many common credibility models. Further, the flexible structure of longitudinal data easily allows actuaries to incorporate many additional characteristics of policyholders into a ratemaking model. In another paper, we show how graphical and diagnostic tools that are available in the longitudinal data setting can be used to specify credibility ratemaking models (Frees et al., 1998).

Appendix A. Best linear unbiased predictor

To check that

$$W_{\text{BLUP}} = (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \mathbf{b}_{\text{GLS}} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y}$$

is the best linear unbiased predictor, consider all other unbiased linear estimators of the form $W_{\text{BLUP}} + \mathbf{c}^t \mathbf{y}$, where $\mathbf{c}$ is a vector of constants. By the unbiasedness, we have that

$$E\mathbf{c}^t \mathbf{y} = EW - EW_{\text{BLUP}} = 0.$$

Thus, $\mathbf{c}^t \mathbf{y}$ is an unbiased estimator of 0. Following Harville (1985), we require this of all possible distributions so that a necessary and sufficient condition for $E\mathbf{c}^t \mathbf{y} = 0$ is $\mathbf{c}^t \mathbf{X} = 0$.

We wish to minimize the mean square prediction error over all choices of $\mathbf{c}$, so consider $E(W_{\text{BLUP}} + \mathbf{c}^t \mathbf{y} - W)^2$. Now,

$$\begin{aligned} \text{Cov}(W_{\text{BLUP}} - W, \mathbf{c}^t \mathbf{y}) &= \text{Cov}((\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \mathbf{b}_{\text{GLS}}, \mathbf{c}^t \mathbf{y}) + \text{Cov}(\text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y}, \mathbf{c}^t \mathbf{y}) - \text{Cov}(W, \mathbf{c}^t \mathbf{y}) \\ &= (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \text{Cov}(\mathbf{b}_{\text{GLS}}, \mathbf{y}) \mathbf{c} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \text{Cov}(\mathbf{y}, \mathbf{y}) \mathbf{c} - \text{Cov}(W, \mathbf{y}) \mathbf{c} \\ &= (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) (\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^t \mathbf{V}^{-1} \text{Cov}(\mathbf{y}, \mathbf{y}) \mathbf{c} + \text{Cov}_{W\mathbf{y}} \mathbf{c} - \text{Cov}_{W\mathbf{y}} \mathbf{c} \\ &= (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) (\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^t \mathbf{c} \\ &= 0. \end{aligned}$$

The last equality follows from $\mathbf{c}^t \mathbf{X} = 0$. Thus, we have

$$E(W_{\text{BLUP}} + \mathbf{c}^t \mathbf{y} - W)^2 = \text{Var}(W_{\text{BLUP}} - W) + \text{Var}(\mathbf{c}^t \mathbf{y}),$$

which can be minimized by choosing $\mathbf{c} = \mathbf{0}$.

Appendix B. Variance of the best linear unbiased predictor

Consider

$$\begin{aligned} \text{Var}(W_{\text{BLUP}} - W) &= \text{Var}((\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \mathbf{b}_{\text{GLS}} + \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y} - W) \\ &= \text{Var}((\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) \mathbf{b}_{\text{GLS}}) + \text{Var}(\text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{y} - W) \\ &= (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X}) (\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1} (\lambda^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \mathbf{X})^t - \text{Cov}_{W\mathbf{y}} \mathbf{V}^{-1} \text{Cov}_{W\mathbf{y}}^t + \sigma_W^2, \end{aligned}$$

as required.

Appendix C. Independence of residuals and least squares estimators

Assume that $\mathbf{y}$ has a multivariate normal distribution with mean $\mathbf{X}\boldsymbol{\beta}$ and variance–covariance matrix $\mathbf{V}$, where $\mathbf{X}$ has dimension $N \times p$ with rank p . Recall that the matrix $\mathbf{V}$ depends on the parameters $\boldsymbol{\tau}$.

We use the matrix $\mathbf{Q} = \mathbf{I} - \mathbf{X}(\mathbf{X}^t \mathbf{X})^{-1} \mathbf{X}^t$. Because $\mathbf{Q}$ is idempotent and has rank $N - p$, we can find an $N \times (N - p)$ matrix $\mathbf{A}$ such that

$$\mathbf{A}\mathbf{A}^t = \mathbf{Q} \quad \text{and} \quad \mathbf{A}^t \mathbf{A} = \mathbf{I}_{N-p}.$$

We also need $\mathbf{G} = \mathbf{V}^{-1} \mathbf{X}(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1}$, an $N \times p$ matrix. Note that $\mathbf{G}^t \mathbf{y} = \mathbf{b}_{\text{GLS}}$, the generalized least squares estimator of $\boldsymbol{\beta}$.

With these two matrices, define the transformation matrix $\mathbf{H} = (\mathbf{A} \mathbf{G})$, an $N \times N$ matrix. Consider the transformed variables

$$\mathbf{H}'\mathbf{y} = \begin{bmatrix} \mathbf{A}'\mathbf{y} \\ \mathbf{G}'\mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{A}'\mathbf{y} \\ \mathbf{b}_{\text{GLS}} \end{bmatrix}.$$

Basic calculations show that

$$\mathbf{A}'\mathbf{y} \sim N(\mathbf{0}, \mathbf{A}'\mathbf{V}\mathbf{A}) \quad \text{and} \quad \mathbf{G}'\mathbf{y} = \mathbf{b}_{\text{GLS}} \sim N(\boldsymbol{\beta}, (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}),$$

in which $\mathbf{z} \sim N(\boldsymbol{\mu}, \mathbf{V})$ denotes that a random vector $\mathbf{z}$ has a multivariate normal distribution with mean $\boldsymbol{\mu}$ and variance–covariance matrix $\mathbf{V}$. Further, we have that $\mathbf{A}'\mathbf{y}$ and $\mathbf{b}_{\text{GLS}}$ are independent. This is due to normality and zero covariance matrix

$$\text{Cov}(\mathbf{A}'\mathbf{y}, \mathbf{b}_{\text{GLS}}) = E(\mathbf{A}'\mathbf{y}\mathbf{y}'\mathbf{G}) = \mathbf{A}'\mathbf{V}\mathbf{G} = \mathbf{A}'\mathbf{X}(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1} = \mathbf{0}.$$

We have $\mathbf{A}'\mathbf{X} = \mathbf{0}$ because $\mathbf{A}'\mathbf{X} = (\mathbf{A}'\mathbf{A})\mathbf{A}'\mathbf{X} = \mathbf{A}'\mathbf{Q}\mathbf{X}$ and $\mathbf{Q}\mathbf{X} = \mathbf{0}$. Zero covariance, together with normality, imply independence.

Appendix D. Restricted likelihoods

To develop the restricted likelihood, we first check the rank of the transformation matrix $\mathbf{H}$. To this end, recall a basic result on determinants of partitioned matrices,

$$\det(\mathbf{M}) = \det \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{M}_{21} & \mathbf{M}_{22} \end{bmatrix} = \det(\mathbf{M}_{11})\det(\mathbf{M}_{22} - \mathbf{M}_{21}\mathbf{M}_{11}^{-1}\mathbf{M}_{12}).$$

See, for example, Graybill (1969) (p. 183). Thus, with $\mathbf{H}$ as in Appendix C,

$$\begin{aligned} \det(\mathbf{H}^2) &= \det(\mathbf{H}'\mathbf{H}) = \det \left[\begin{bmatrix} \mathbf{A}' \\ \mathbf{G}' \end{bmatrix} [\mathbf{A} \mathbf{G}] \right] = \det \begin{bmatrix} \mathbf{A}'\mathbf{A} & \mathbf{A}'\mathbf{G} \\ \mathbf{G}'\mathbf{A} & \mathbf{G}'\mathbf{G} \end{bmatrix} = \det(\mathbf{A}'\mathbf{A})\det(\mathbf{G}'\mathbf{G} - \mathbf{G}'\mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'\mathbf{G}) \\ &= \det(\mathbf{G}'\mathbf{G} - \mathbf{G}'\mathbf{Q}\mathbf{G}) = \det(\mathbf{G}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{G}) = \det((\mathbf{X}'\mathbf{X})^{-1}), \end{aligned}$$

using $\mathbf{G}'\mathbf{X} = \mathbf{I}$. Thus, the transformation $\mathbf{H}$ is non-singular if and only if $\mathbf{X}'\mathbf{X}$ is non-singular. In this case, no information is lost by considering the transformation $\mathbf{H}'\mathbf{y}$.

We now develop the restricted likelihood based on the probability density function of $\mathbf{A}'\mathbf{y}$. We first note a relationship used by Harville (1974), concerning the probability density function of $\mathbf{G}'\mathbf{y}$. We write $f_{\mathbf{G}'\mathbf{y}}(\mathbf{z}, \boldsymbol{\beta})$ to denote the probability density function of the random vector $\mathbf{G}'\mathbf{y}$, evaluated at the (vector) point $\mathbf{z}$ with mean (vector) parameter $\boldsymbol{\beta}$. Because probability density functions integrate to 1, we have the relation

$$\begin{aligned} 1 &= \int f_{\mathbf{G}'\mathbf{y}}(\mathbf{z}, \boldsymbol{\beta}) \, d\mathbf{z} = \int \frac{1}{(2\pi)^{p/2}\det(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1/2}} \exp\left(-\frac{1}{2}(\mathbf{z} - \boldsymbol{\beta})'\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}(\mathbf{z} - \boldsymbol{\beta})\right) d\mathbf{z} \\ &= \int \frac{1}{(2\pi)^{p/2}\det(\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1/2}} \exp\left(-\frac{1}{2}(\mathbf{z} - \boldsymbol{\beta})'\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}(\mathbf{z} - \boldsymbol{\beta})\right) d\boldsymbol{\beta} = \int f_{\mathbf{G}'\mathbf{y}}(\mathbf{z}, \boldsymbol{\beta}) \, d\boldsymbol{\beta}, \quad \text{for each } \mathbf{z}, \end{aligned}$$

with a change of variables.

Because of the independence of $\mathbf{A}'\mathbf{y}$ and $\mathbf{G}'\mathbf{y} = \mathbf{b}_{\text{GLS}}$, we have $f_{\mathbf{H}'\mathbf{y}} = f_{\mathbf{A}'\mathbf{y}}f_{\mathbf{G}'\mathbf{y}}$. Here, $f_{\mathbf{H}'\mathbf{y}}$, $f_{\mathbf{A}'\mathbf{y}}$ and $f_{\mathbf{G}'\mathbf{y}}$ are the density functions of the random vectors $\mathbf{H}'\mathbf{y}$, $\mathbf{A}'\mathbf{y}$ and $\mathbf{G}'\mathbf{y}$, respectively. For notation, let $\mathbf{y}^*$ be a potential realization of the random vector $\mathbf{y}$. Thus, the probability density function of $\mathbf{A}'\mathbf{y}$ is

$$f_{\mathbf{A}'\mathbf{y}}(\mathbf{A}'\mathbf{y}^*) = \int f_{\mathbf{A}'\mathbf{y}}(\mathbf{A}'\mathbf{y}^*)f_{\mathbf{G}'\mathbf{y}}(\mathbf{G}'\mathbf{y}^*, \boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int f_{\mathbf{H}'\mathbf{y}}(\mathbf{H}'\mathbf{y}^*, \boldsymbol{\beta}) \, d\boldsymbol{\beta} = \int \det(\mathbf{H})^{-1} f_{\mathbf{y}}(\mathbf{y}^*, \boldsymbol{\beta}) \, d\boldsymbol{\beta},$$

using a change of variables. Now, let $\mathbf{b}_{\text{GLS}}^*$ be the realization of $\mathbf{b}_{\text{GLS}}$ using $\mathbf{y}^*$. Then, from a standard equality from analysis of variance,

$$(\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta})^t \mathbf{V}^{-1}(\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}) = (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*)^t \mathbf{V}^{-1}(\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*) + (\mathbf{b}_{\text{GLS}}^* - \boldsymbol{\beta})^t \mathbf{X}^t \mathbf{V}^{-1} \mathbf{X}(\mathbf{b}_{\text{GLS}}^* - \boldsymbol{\beta}).$$

With this equality, the probability density function $f_{\mathbf{y}}$ can be expressed as

$$\begin{aligned} f_{\mathbf{y}}(\mathbf{y}^*, \boldsymbol{\beta}) &= \frac{1}{(2\pi)^{N/2} \det(\mathbf{V})^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta})^t \mathbf{V}^{-1} (\mathbf{y}^* - \mathbf{X}\boldsymbol{\beta}) \right) \\ &= \frac{1}{(2\pi)^{N/2} \det(\mathbf{V})^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*)^t \mathbf{V}^{-1} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*) \right) \\ &\quad \times \exp \left(-\frac{1}{2} (\mathbf{b}_{\text{GLS}}^* - \boldsymbol{\beta})^t \mathbf{X}^t \mathbf{V}^{-1} \mathbf{X} (\mathbf{b}_{\text{GLS}}^* - \boldsymbol{\beta}) \right) \\ &= \frac{(2\pi)^{p/2} \det(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1/2}}{(2\pi)^{N/2} \det(\mathbf{V})^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*)^t \mathbf{V}^{-1} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*) \right) f_{\mathbf{G}^t \mathbf{y}}(\mathbf{b}_{\text{GLS}}^*, \boldsymbol{\beta}). \end{aligned}$$

Thus,

$$\begin{aligned} f_{\mathbf{A}^t \mathbf{y}}(\mathbf{A}^t \mathbf{y}^*) &= \frac{(2\pi)^{p/2} \det(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1/2}}{(2\pi)^{N/2} \det(\mathbf{V})^{1/2}} \det(\mathbf{H})^{-1} \exp \left(-\frac{1}{2} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*)^t \mathbf{V}^{-1} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*) \right) \\ &\quad \times \int f_{\mathbf{G}^t \mathbf{y}}(\mathbf{b}_{\text{GLS}}^*, \boldsymbol{\beta}) \, d\boldsymbol{\beta} \\ &= (2\pi)^{-(N-p)/2} \det(\mathbf{V})^{-1/2} \det(\mathbf{X}^t \mathbf{X})^{1/2} \det(\mathbf{X}^t \mathbf{V}^{-1} \mathbf{X})^{-1/2} \\ &\quad \times \exp \left(\frac{1}{2} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*)^t \mathbf{V}^{-1} (\mathbf{y}^* - \mathbf{X}\mathbf{b}_{\text{GLS}}^*) \right). \end{aligned}$$

This yields the REML likelihood in Section 4.2, after taking logarithms and dropping constants that do not involve $\boldsymbol{\tau}$.

References

- Baltagi, B.H., 1995. *Econometric Analysis of Panel Data*. Wiley, New York.
- Bühlmann, H., 1967. Experience rating and credibility. *ASTIN Bulletin* 4, 199–207.
- Bühlmann, H., Gisler, A., 1997. Credibility in the regression case revisited. *ASTIN Bulletin* 27, 83–98.
- Bühlmann, H., Straub, E., 1970. Glaubwürdigkeit für Schadensätze. *Mitteilungen der Vereinigung Schweizerischer Versicherungsmathematiker* 70, 111–133.
- Corbeil, R.R., Searle, S.R., 1976a. Restricted maximum likelihood (REML) estimation of variance components in the mixed model. *Technometrics* 18, 31–38.
- Corbeil, R.R., Searle, S.R., 1976b. A comparison of variance components estimators. *Biometrics* 32, 779–791.
- Dannenburg, D.R., 1994. Some results on the estimation of the credibility factor in the classical Bühlmann model. *Insurance: Mathematics & Economics* 14, 39–50.
- Dannenburg, D.R., Kaas, R., Goovaerts, M.J., 1996. *Practical Actuarial Credibility Models*. Institute of Actuarial Science and Econometrics, University of Amsterdam, Amsterdam, The Netherlands.
- De Vylder, F., 1981. Practical credibility theory with emphasis on optimal parameter estimation. *ASTIN Bulletin* 12, 115–131.
- Diggle, P.J., Liang, K.-Y., Zeger, S.L., 1994. *Analysis of Longitudinal Data*. Clarendon Press, Oxford.
- Frees, E. W., Young, V.R., Luo, Y., 1998. Credibility ratemaking using longitudinal data models, working paper, University of Wisconsin-Madison.
- Goldberger, A.S., 1962. Best linear unbiased prediction in the generalized linear regression model. *Journal of the American Statistical Association* 57, 369–375.
- Goovaerts, M. J., Hoogstad, W., 1987. *Credibility Theory, Surveys of Actuarial Studies*. Nationale-Nederlanden N. V., Rotterdam.
- Graybill, F.A., 1969. *Matrices with Applications in Statistics*, 2nd ed. Wadsworth, Belmont, California.

- Hachemeister, C.A., 1975. Credibility for regression models with applications to trend. In: Kahn, P.M. (Ed.), *Credibility: Theory and Applications*. Academic Press, New York, pp. 129–163.
- Hand, D.J., Crowder, M.J., 1996. *Practical Longitudinal Data Analysis*. Chapman-Hall, London.
- Harville, D.A., 1974. Bayesian inference for variance components using only error contrasts. *Biometrika* 61, 383–385.
- Harville, D.A., 1977. Maximum likelihood approaches to variance component estimation and to related problems, with discussion. *Journal of the American Statistical Association* 72, 320–340.
- Harville, D.A., 1985. Decomposition of prediction error. *Journal of the American Statistical Association* 80, 132–138.
- Henderson, C.R., 1973. Sire evaluation and genetic trends, in *Proceedings of the Animal Breeding and Genetics Symposium in Honor of Dr. Jay L. Lush*, 10–41. Amer. Soc. Animal Sci.-Amer. Dairy Sci. Assn. Poultry Sci. Assn., Champaign, IL.
- Hsiao, C., 1986. *Analysis of Panel Data*. Cambridge University Press, Cambridge.
- Jewell, W.S., 1975. The use of collateral data in credibility theory: A hierarchical model, *Giornale dell'Istituto Italiano degli Attuari*, 38, 1–16. (Also, Research Memorandum 75-24 of the International Institute for Applied Systems Analysis, Laxenburg, Austria.)
- Kaas, R., Dannenburg, D., Goovaerts, M., 1997. Exact credibility for weighted observations, *ASTIN Bulletin* 27, 287–296.
- Klugman, S., 1992. *Bayesian Statistics in Actuarial Science*. Kluwer, Boston.
- Klugman, S., Panjer, H., Willmot, G., 1998. *Loss Models: From Data to Decisions*. Wiley, New York.
- Ledolter, J., Klugman, S., Lee, C.-S., 1991. Credibility models with time-varying trend components. *ASTIN Bulletin* 21, 73–91.
- Lee, Y., Nelder, J.A., 1996. Hierarchical generalized linear models. *J. R. Stat. Soc., Series B* 58, 619–678.
- Lehmann, E., 1991. *Theory of Point Estimation*. Wadsworth & Brooks/Cole, Pacific Grove, CA.
- Mátyás, L., Sevestre, P. (Eds.), 1996. *The Econometrics of Panel Data*. Kluwer, Dordrecht.
- McCullagh, P., Nelder, J., 1989. *Generalized Linear Models*, 2nd ed. Chapman and Hall, London.
- Nelder, J.A., Verrall, R.J., 1997. Credibility theory and generalized linear models. *ASTIN Bulletin* 27, 71–82.
- Norberg, R., 1980. Empirical Bayes credibility. *Scandinavian Actuarial Journal* 1980, 177–194.
- Norberg, R., 1982. On optimal parameter estimation in credibility. *Insurance: Mathematics & Economics* 1, 73–89.
- Norberg, R., 1986. Hierarchical credibility: Analysis of a random effect linear model with nested classification, *Scandinavian Actuarial Journal*, 204–222.
- Patterson, H.D., Thompson, R., 1971. Recovery of inter-block information when block sizes are unequal. *Biometrika* 58, 545–554.
- Potthoff, R.F., Roy, S.N., 1964. A generalized multivariate analysis of variance model useful especially for growth curve problems. *Biometrika*, 51, 313–326.
- Rao, C.R., 1970. Estimation of variance and covariance components in linear models. *Journal of the American Statistical Association* 67, 112–115.
- Robinson, G.K., 1991. That BLUP is a good thing: The estimation of random effects, with discussion. *Statistical Science* 6, 15–51.
- Swallow, W.H., Searle, S.R., 1978. Minimum variance quadratic unbiased estimation (MIVQUE) of variance components. *Technometrics* 20, 265–272.
- Swamy, P.A.V.B., 1970. Efficient inference in a random coefficient regression model. *Econometrica* 38, 311–323.
- Wolfinger, R., Tobias, R., Sall, J., 1994. Computing Gaussian likelihoods and their derivatives for general linear mixed models. *SIAM Journal of Scientific Computing* 15 (6) 1294–1310.